

Lab 3 – Component Modelling & Architectural Pattern Selection

Written Justification – Academic Elective Bidding & Allocation System (Problem Statement #05)

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1. Architectural Choice

We chose the **Microservices Architecture** for the Academic Elective Bidding & Allocation System (Problem Statement #05). The system is decomposed into nine components, each owning one requirement area: Web Portal; API Gateway & Auth Service; Bid Manager Service (FR-001); Credit & Eligibility Service (FR-001 credit ledger and prerequisite validation); Catalog & Config Service (FR-002 bidding window and seat capacity); Allocation Engine Service (FR-003 collision-free allocation); Swap & Withdrawal Service (FR-005); Notification Service (FR-004); and the Bid & Allocation Datastore.

2. Two Scenario-Specific Reasons

Reason 1 - The two heaviest requirements have opposite scaling profiles. NFR-002 demands at least 500 students submitting bids concurrently across a short window, which is a high-volume, short-transaction write load on the Bid Manager Service. NFR-001 demands that the solver process 5,000 bids and resolve timetable collisions in under 30 seconds, which is a single CPU-bound batch job in the Allocation Engine that runs once per cycle. Microservices let us replicate Bid Manager horizontally while the window is open and give the Allocation Engine dedicated compute only at cut-off, instead of sizing one monolith for both.

Reason 2 - The requirements change on different clocks and must fail independently. FR-004 depends on an external mail provider; in a monolith a slow provider would hold request threads and degrade bidding, whereas an asynchronous Notification Service cannot block FR-001 submissions, which directly protects the 99.5% availability in NFR-002. Similarly, FR-005 swap and withdrawal handling is only active after allocation and FR-002 configuration is only touched by the Registrar between cycles, so each can be redeployed on its own schedule without a full-system release during a live bidding window.

3. Security Advantage

All external traffic terminates at one API Gateway & Auth Service, giving a single enforcement point for TLS, authentication and role-based authorisation - which is what the security half of NFR-001 requires. The two actors are separated at that boundary: Registrar-only operations (IConfigAdmin for the FR-002 window and capacity, IAllocationRun for FR-003, and the approval step in FR-005) are unreachable from a student session. Credit balances and prerequisite checks sit behind IEligibilityCheck in their own service, so a tampered client cannot exceed the 100-credit limit or bypass a prerequisite (FR-001), and the Bid & Allocation Datastore is reachable only through the Bid Manager Service's IBidRepository interface, so the public-facing portal has no direct database access and a compromised front end has a limited blast radius.

4. Performance Benefit

Because the Bid Manager Service is stateless, extra instances can run behind the gateway only for the days the FR-002 window is open, so the 500-user peak in NFR-002 is spread across replicas instead of queueing behind one process. The Allocation Engine then runs as a dedicated compute-bound service that loads the full bid set into memory without competing with portal traffic, which is what makes the sub-30-second target for 5,000 bids in NFR-001 achievable. FR-004 notification is published over INotifyResult asynchronously, so e-mail delivery latency sits outside the allocation critical path and the 24-hour notification guarantee does not delay result publication.

5. Trade-off Acknowledged

Microservices add operational complexity and eventual consistency. We contain this by keeping every bid write inside the Bid Manager Service and its datastore, so the FR-001 credit total stays transactional, and by using asynchronous messaging only where delay is acceptable (FR-004 notifications).

Mapping to the given components: the handout's "Order Manager Component" maps to the Bid Manager Service (it manages the student's submission) and "Payment Service Component" maps to the Credit & Eligibility Service (bidding credits replace money in this scenario). The remaining components identified are the API Gateway & Auth Service, Catalog & Config Service, Allocation Engine Service, Swap & Withdrawal Service, Notification Service, Web Portal and Bid & Allocation Datastore. Interfaces shown: IPortalAPI, IBidSubmission, IConfigAdmin, ICatalogQuery, IEligibilityCheck, IAllocationRun, IBidQuery, IBidRepository, ISwapRequest, ISeatUpdate, INotifyResult.